

Internet of Things (IoT)-Enabled Solar Micro Inverter using Blockchain Technology

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Abstract—Performance of the conventional photovoltaic (PV) systems is limited by the on-board embedded systems in the solar inverters. Moreover, an expensive supervisory control and data acquisition (SCADA) system or phase management units (PMUs) are still required for intelligent maintenance of the PV systems. It is expected that the conventional solar micro inverters can be advanced as a result of further investigation of the emerging Internet of Things (IoT) technologies and cloud/edge computing services. This paper proposes an internet of things (IoT)-enabled solar inverter using **blockchain technology**, which improving connectivity, observability, cyber security, and decentralized control toward a smarter PV system in smart grid environments. The IoT device incorporating blockchain technology enables the solar micro inverter to securely: 1) aggregate and send data to a main server for PV system health monitoring; 2) sharing data with other devices for **decentralized** optimal controls; and 3) execute PV inverter-level energy transaction and record. The concept of the proposed IoT-enabled solar micro inverter is validated by experimental studies.

Keywords— *blockchain, internet of things, IoT, smart contract, solar micro inverter*

I. INTRODUCTION

According to the International Energy Agency, solar energy was the fastest-growing source of global energy in 2017 [1]. To integrate the solar energy source to the electric grid, an inverter is desired [2]. **Grid-tied solar micro inverters for photovoltaic (PV) systems are now getting more attention for their individual maximum power point tracking (MPPT) ability, high efficiency, high power density, voltage and power quality and reliability than string inverters and a centralized inverter [3].**

The advent of Internet of Things (IoT) with cloud or edge computing supports is expected to advance the PV systems toward smart PV systems by **fully utilizing IoT network**, powerful computing and resources such as artificial intelligent (AI) analytics and visualization tools, resulting in providing significant value in cost reduction, extended scalability, greater visibility, and optimal control in smart grid environments. The

IoT devices integrated in the PV inverters enable power monitoring [4], health monitoring [5] services in cloud, and further controlling individual solar micro inverters via Internet. Moreover, the concept of IoT-enabled micro inverters support machine-to-machine communications leveraging decentralized optimal controls [6] and peer to peer (P2P) electricity trading [7] in smart grid environments.

However, there are critical concerns and challenges about cybersecurity of the IoT-enabled PV systems from cyber-attacks since the IoT-embedded solar micro inverters are always connected to internet through the IoT devices. Cyber-attacks targeting the solar micro inverters will impose new security and safety risks, specifically, maliciously intending to damage or disable PV systems [8], [9]. Changing any parameters of the control algorithm in the inverter will result in dangerous impacts on the PV systems. Furthermore, the solar micro inverters with malware injected by hackers will be botnets that enable attacks other devices sharing networks in smart grid [10]. In addition, a cloud service from the trusted party is still vulnerable to a single point of failure from cyber-attacks and the challenge of longevity after vendors stop maintaining the cloud service [11]. With an awareness of these security concerns and challenges, investigation of the cybersecurity vulnerabilities (i.e., weaknesses) and guidance for mitigating cyber-attacks is imminently required to leverage the proliferation of the IoT-enabled PV systems.

Blockchain is a distributed database that maintains a continuously growing list of data records secured from tampering and revision [12]. Recently, blockchain technology incorporating blockchain ledgers and smart contracts which are self-executing script has been widely studied in many applications such as peer-to-peer (P2P) transaction, supply chain, energy trading using PV systems [7], demand-side management [13], and IoT security and privacy [14]. However, the investigation of cybersecurity for IoT-enabled solar micro inverter in cyber-physical environments using blockchain technology has been less studied.

The goal of this paper is to propose the IoT-enabled solar micro inverters using blockchain technology toward the smarter PV systems. The proposed blockchain-IoT enabled solar micro inverter can securely: 1) aggregate and send data to a cloud server for PV system health monitoring; 2) communicate with other devices for a decentralized cooperative control; and 3) execute solar microgrid-level energy transactions and records. The proposed IoT-enabled inverter is built using a TI's solar micro inverter and an IoT device and a smart contract is

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implemented in the IoT device on Hyper-ledger Fabric blockchain platform. Experimental results validate the concept of the proposed IoT-enabled inverter.

II. RELATED WORK

A. Blockchain Technology

Blockchain Technology (e.g., Bitcoin [15] and Ethereum [16]) is an emerging technology used for secured transactions/database and network, which is a combination of trust mechanisms (e.g., distributed database and cryptography), a consensus algorithm, and smart contracts [16]. Blockchain is a distributed data structure consisting of timestamped blocks and links between blocks called “chain” and the blocks are inherently resistant to tampering and revision [12]. The general blockchain structure used in Bitcoin and Ethereum is shown in Fig. 1. A block consists of a block header and a body. The block header mainly includes block ID, a timestamp, a hash of the previous block (i.e., a cryptographic link creating a chain and tamper-proof), a random nonce (i.e., used for solving the proof of work (PoW)), and Merkle tree root (i.e., encoded transactions/data in the block in a single hash for efficient data verification; any data modification will change the hash value so that it is easy to check data integrity). The types of the transaction will include records of the transfer of assets or data, broadcast messages, and smart contracts, which are encrypted by cryptographic digital signatures (e.g., users’ private keys). Only participants who have the cryptographic keys can verify the data, time, and user of the transaction (i.e., data privacy). Therefore, such cryptographic methods will bring data integrity, privacy, and authentication.

Smart contracts are self-executing scripts that execute the terms of contracts triggered by designated events. If all the conditions of the contracts are satisfied, the blockchain network will execute the contract terms automatically and independently in a prescribed manner. Because a smart contract with a unique address is stored on the blockchain, users or nodes in the blockchain network can trigger the smart contract by addressing the transaction. Therefore, users have a capability of designing and implementing codes in the form of smart contracts for automated and efficient trading or workflows since smart contract provides an interface between the blockchain network

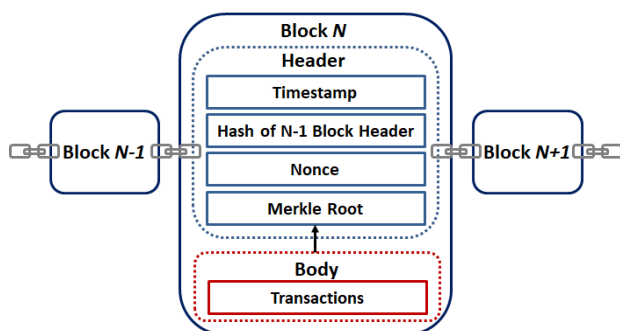


Fig. 1. General blockchain structure.

and the physical world.

In view of the fact that the blockchain is hosted, updated, validated by individual peer nodes rather than by a single centralized authority, the blockchain improves the trust, security, and transparency of transactions/data due to inherent benefits such as immutability, auditability, data integrity and authentication, fault tolerance, and above all trust-free operation [12]. Moreover, the idea blockchain theory has brought about a potential solution to the IoT security problems [14].

B. Overall Architecture of the Proposed Cyber-Physical PV System

Fig. 2 shows an overall system architecture of the proposed cyber-physical PV system consisting of an IoT-enabled PV system, a cloud-based PV management platform, and a blockchain server in smart grid environments. The IoT-enabled PV system includes IoT-enabled solar micro inverters, where the IoT device interfacing with the solar micro inverter acts as a participant node in the blockchain network and blockchain ledgers and smart contracts are implemented in the IoT device. The IoT device is a key in the IoT-enabled inverter which contains an IoT system-on-chip and a communication component (a short-range radio network, e.g., Wi-Fi, ZigBee, BLE) to communicate with other inverters and other systems (e.g., battery systems and energy management system (EMS)) using an IoT Gateway and/or 4G/5G network connected to the cloud support server via internet. The blockchain technology is used to further secure communication and data on top of the IoT network.

The cloud-based PV management platform is designed to provide cloud supports including cloud-based PV power management, optimal control, health monitoring and energy trading, etc. The blockchain server consisting of miners and

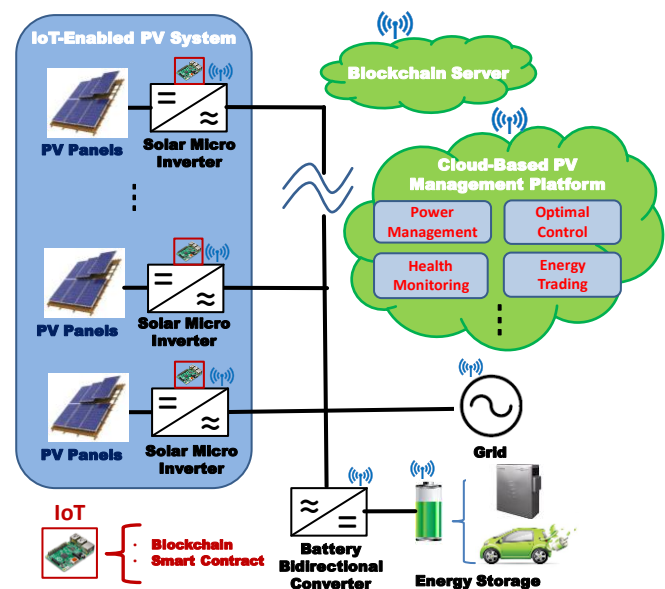


Fig. 2. The proposed cyber-physical PV system consisting of IoT-enabled solar micro inverters, a cloud-based PV management platform, and blockchain server in smart grid environments.

validation nodes provides blockchain services such as transaction/data validation, mining, ID management, and copy of blockchain.

III. THE PROPOSED IoT-ENABLED SOLAR MICRO INVERTER

This paper focuses on proposing an IoT-enabled solar micro inverter using blockchain technology. Fig. 3 shows an architecture of blockchain technology implemented in the IoT nodes in the blockchain network. With blockchain technology, the IoT peer node can perform: 1) an interface that extracts data from the inverter (i.e., data aggregation) or controlling the inverter (i.e., remote control); 2) communication with other nodes and cloud servers in the blockchain-based IoT network (e.g., private encrypted channels and blockchain ledger); 3) store the shared blockchain ledgers; and 4) execute smart contracts. Therefore, it is expected that the IoT enabled micro solar inverter will be enhanced and enables: 1) sending secured data to a cloud server for PV system health monitoring; 2) communication with other devices for decentralized optimal controls securely; and 3) executing the energy transaction in PV inverter/system level. In this paper, we propose the use of the IBM's Hyperledger-Fabric [17] that will be more applicable to the IoT applications than other BC platforms (e.g., Ethereum and IoTA) since the Hyperledger-Fabric provides private and permissioned BC, requires less energy and computational requirements in consensus protocols, and does not require transaction fees/coins [18].

A. Interfacing an IoT Device to a Solar Micro Inverter

Fig. 4 shows the topology and control structure of the IoT-enabled solar micro inverter which is built by using a 250-W TI's solar micro inverter [19] and a LattePanda IoT device [20]. The two-stage micro inverter consists of a DC-DC flyback converter and a DC-AC inverter. The function of the DC-DC flyback converter is to step up the low-level PV voltage (V_{pv}) to a high-level grid/load voltage (V_g) required by the cascaded DC-AC inverter, which regulates voltage from DC to AC and controls the DC bus voltage (V_{bus}) to be constant. For the DC/DC converter, reference current I_{pv_ref} is provided by an MPPT controller. A Piccolo TMS302F28035 Isolated control card collects the data, executes the MPPT algorithm, and

generates PWM signals for the inverter. The control card is interconnected by the external IoT device so that the IoT device

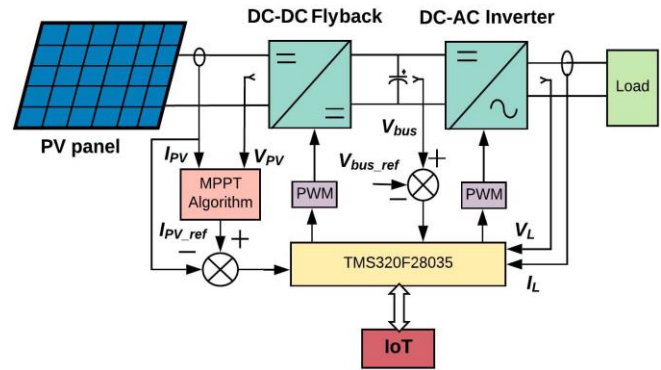


Fig. 4. Topology and control structure of a single-phase IoT-enabled solar micro inverter.

can access data stored in the control card and control the inverter.

B. Blockchain Implementation Using Hyperledger-Fabric

As shown in Fig. 3, blockchain and smart contracts are implemented in the blockchain network which is constituent of IoT client nodes and a blockchain server called "Fabric" using the Hyperledger-Fabric platform. The IoT client node executes smart contract called "chaincode" in Hyperledger-Fabric. After enrolling the client via Fabric-CA (i.e., membership service) and a certificate is issued to the client. Then, the authorized client can communicate with a peer node in the blockchain server using Fabric-SDK-node.

A block is made as follows. First, the IoT-enabled micro inverter sends data to the blockchain server using encrypted Rest API. The data is then sent to the endorsement peers to validate the data with the chaincode. After validation, the data is

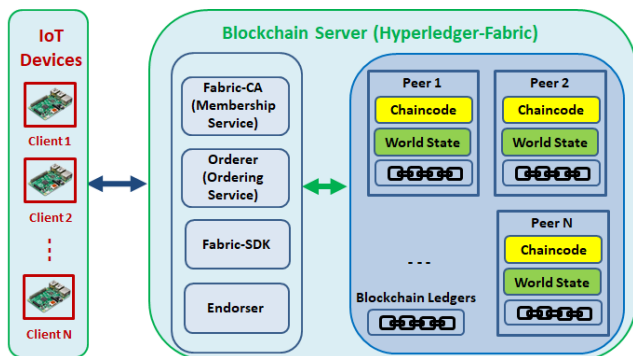


Fig. 3. Architecture of combining Blockchain and smart contract in the IoT-enabled solar micro inverters.

* See the License for the specific language governing permissions and limitations under the License.
*/

```
namespace org.example.biznet {
    participant PowerParticipant identified by participantId {
        o String participantId
        o String firstName
        o String lastName
    }

    asset PVInfo identified by valuenumber {
        o String valuenumber
        o String PVId
        --> PowerParticipant owner
        o String current
        o String voltage
        o String power
    }

    transaction PowerTransaction {
        --> PVInfo PV
        o String newValue
    }

    event PVEvent {
        --> PVInfo PV
        o String oldValue
        o String newValue
    }
}
```

Fig. 5. An example chaincode.

sent to an orderer peer that sequences the data into a block. The block is sent to peers.

Chain codes are designed for the solar micro inverters to: 1) broadcast messages/commands and exchange data through the blockchain ledgers; 2) exchange data which is not required to store in the blockchain ledgers through an encrypted private IoT network in the PV systems; and 3) store and read the blockchain ledgers. Therefore, the proposed blockchain-based framework can improve the communication and data security of the IoT-enabled solar micro inverters. Chaincode is deployed on both the server and the IoT devices to access and create the blockchain ledgers. The chaincode can be written on Node.js, go and Java language. This chain code will be controlled by the Hyperledger Composer Rest Server that will be installed on the blockchain server. Fig. 5 shows an example chaincode that stores the PV data in the blockchain ledgers.

IV. EXPERIMENTAL RESULTS

Fig. 6 illustrates the experimental setup consisting of an IoT-enabled solar micro inverter, a PV simulator (i.e., DC power supply), a resistive load, an IoT device for an energy management system (EMS), and the blockchain server built in a virtual cloud server (cloud.ibm.com). As described in Section III, The LattePanda IoT device is directly connected to a Piccolo-B control card in the solar micro inverter and collects necessary information (e.g., V_{pv} , I_{pv} , V_L , I_L , V_{bus} and generated power P_{pv}). The Hyperledger-Fabric platform is implemented into the two IoT devices and the server acting as client nodes and a blockchain server node in the blockchain network, respectively. Chain codes are written and implemented into the nodes.

First, we validate a case of broadcasting PV data from the IoT-enabled micro inverter to the EMS through the BC ledger. Fig. 7 shows the PV voltages and currents in Data-1 and Data-2 recorded in the blockchain ledger stored in the EMS node. Therefore, the EMS can know the status of PV system by reading data in the blockchain ledger. It is observed that it takes about 3 seconds until the EMS can see the PV data after the

IoT-enabled micro inverter sent the data. This latency time will be further shorter if the blockchain server is locally installed.

Fig. 8 illustrates the oscilloscope waveforms of V_L and I_L that are measured from the inverter board and Fig. 9 shows the

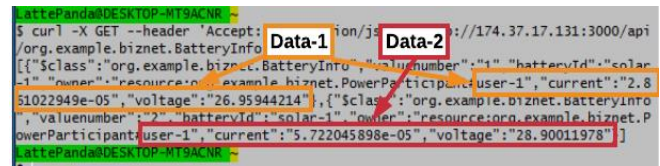


Fig. 7. Screen shot of PV data (V_{pv} and I_{pv}) in the blockchain ledger.

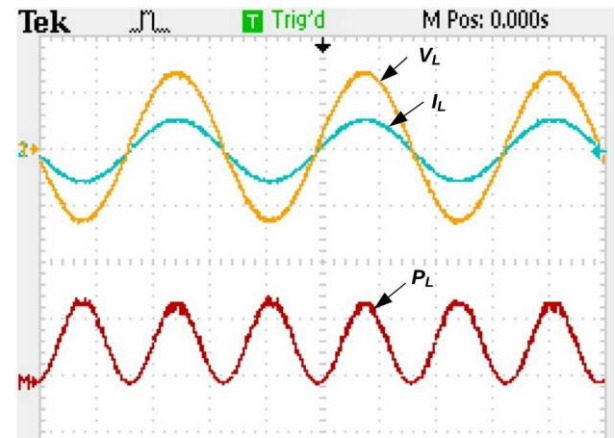


Fig. 8. Screen shot of PV data (V_{pv} and I_{pv}) in the blockchain ledger.

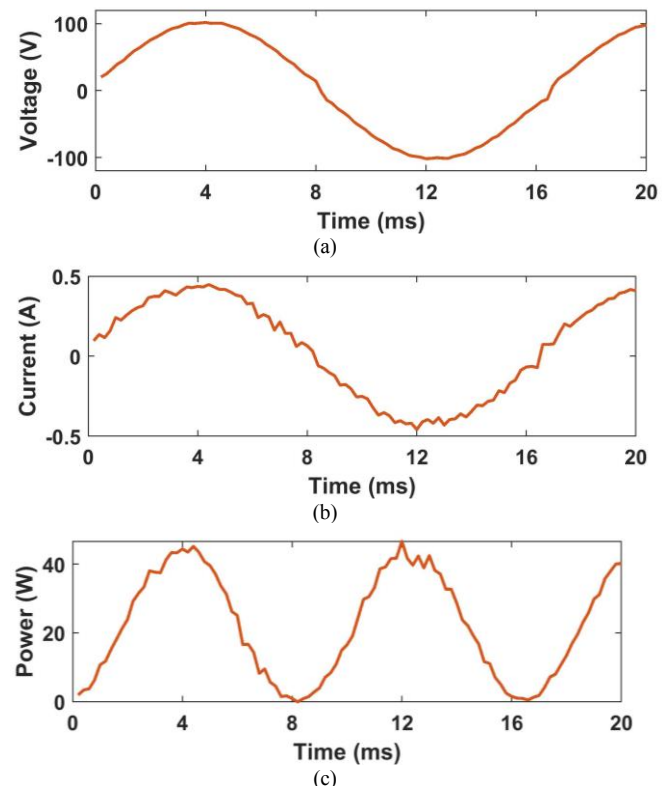


Fig. 9. The waveforms of data extracted from the blockchain ledger: (a) V_L (b) I_L , and (c) P_L .

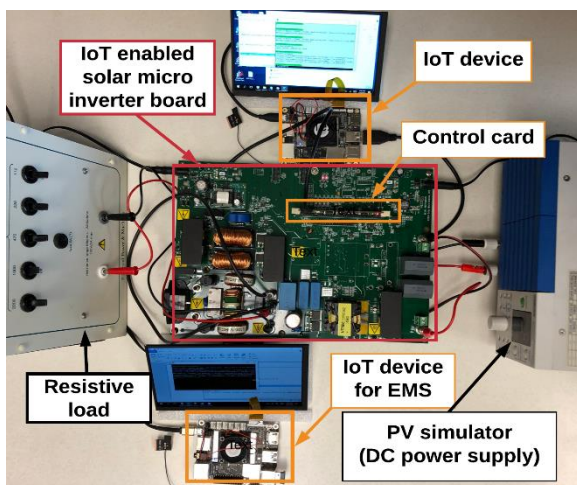


Fig. 6. Experimental setup.



Transaction List for EMS

SL#	ID	Vrms(V)	Irms(mA)	Prms(W)	Vbus(V)
1	INV-1	59	269	16	81
2	INV-1	59	279	16	81
3	INV-1	59	273	16	81
4	INV-1	60	273	16	81
5	INV-1	59	278	16	81
6	INV-1	59	269	16	81
7	INV-1	60	272	16	81
8	INV-1	59	272	16	81
9	INV-1	59	262	16	81

Fig. 10. Web-based user interface for energy transaction with Hyperledger Fabric.

graphical representation of the inverter information extracted from the blockchain ledger. It is obvious that such data can be used for comprehensive PV system health monitoring and management in the cloud-based PV management platform. Furthermore, Fig. 10 shows the blockchain records of the RMS power supplied to the load from the solar micro inverter, which can be potentially used for transaction records.

Therefore, the experimental results open a new opportunities and challenges of the adoption of the blockchain-based communication and data storage in the PV systems.

V. CONCLUSIONS

This paper has introduced the IoT-enabled solar micro inverters using blockchain technology. Through the blockchain technology, The IoT device enables to secure communication and data used in the PV system. The experimental results provide potential opportunities of enhancing cybersecurity of the PV system and energy trade in an inverter/PV system level as well as challenge issues such as growing ledger, reducing complexity and latency, increasing scalability and requiring additional blockchain server. Future works include implementing local blockchain, the private channel and investigation of the cyber-physical security vulnerabilities and guidance for mitigating cyber-attacks through the combination of blockchain technology and artificial intelligence to leverage the proliferation of the IoT-enabled PV systems in smart grid environments.

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